

HP Therm-γ Material Safety Data Sheet

SECTION 1: Product and Company Identification

Product Name : HP Therm-γ

Product Code :

Manufacturer or supplier Details

Company : Hindustan Petroleum Corporation Limited

Address :

Telephone :

Recommended use of the chemical and restrictions on use

Recommended use : Heat transfer fluids

Restrictions on use : Not Known

SECTION 2: Hazards identification

Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008

Acute toxicity (Oral) : Category 4 (H302 – Harmful if inhaled)

Skin irritation : Category 2 (H315 – Causes skin irritation)

Eye irritation : Category 2 (H319 – Causes serious eye irritation)

Specific target organ toxicity : Category 3 (H335 – May cause respiratory irritation)

Acute aquatic toxicity : Category 1 (H400 – Very toxic to aquatic life)

Chronic aquatic toxicity : Category 2 (H411 – Toxic to aquatic life with long lasting effects)

Label elements

Labelling according Regulation (EC) No 1272/2008

Labelling



Signal word : Warning

Hazard statement(s)

H302	Harmful if inhaled
H315	Causes skin irritation
H319	Causes serious eye irritation
H335	May cause respiratory irritation
H400	Very toxic to aquatic life
H411	Toxic to aquatic life with long lasting effects

Precautionary statement(s)

P261	Avoid breathing dust.
P264	Wash skin thoroughly after handling.
P273	Avoid release to the environment
P280	Wear protective gloves
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P302 + P352	IF ON SKIN: Wash with plenty of water
P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P501	Dispose of contents/ container to an approved waste disposal plant

Other hazards which do not result in classification

None known

SECTION 3: Composition/information on ingredients**Components**

Chemical name	CAS-No.	Concentration (Wt. %)
Diphenyl oxide	101-84-8	70-95
Biphenyl	92-52-4	5-30

SECTION 4: First aid measures

Description of first aid measures

If inhaled	• Remove person to fresh air and keep comfortable for breathing. • If breathing is difficult, give oxygen. • Consult a physician if necessary.
In case of skin contact	• Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. • If skin irritation occurs: Get medical advice/ attention. • Wash contaminated clothing before reuse
In case of eye contact	• In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. • Get medical attention if irritation persists
If swallowed	• IF SWALLOWED: Immediately call a POISON CENTER/doctor. • Do NOT induce vomiting. • Rinse mouth. • Never give anything by mouth to an unconscious person
Most important symptoms and effects, both acute and delayed	• May be harmful if swallowed. • Causes skin irritation. • Harmful if inhaled. • May cause respiratory irritation.

SECTION 5: Firefighting measures

Suitable extinguishing media	: Water spray, Carbon dioxide (CO ₂), Dry chemical & alcohol-resistant foam
Unsuitable extinguishing media	: Do not use a solid water stream as it may scatter and spread fire.

Special hazards arising from the substance or mixture	: Carbon oxides
Specific extinguishing methods	: Use a water spray to cool fully closed containers
Special protective equipment for firefighter	: Wear an approved positive pressure self-contained breathing apparatus in addition to standard firefighting gear

SECTION 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures	• Use personal protective equipment. • Avoid dust formation. • Avoid breathing vapors, mist or gas. • Ensure adequate ventilation. • Evacuate personnel to safe areas. • Avoid breathing dust. • Local authorities should be advised if significant spillages cannot be contained
Environmental precautions	• Clear up spills immediately and dispose of waste safely. • Avoid release to the environment. • Do not let product enter drains, sewers or streams • Collect spillage
Methods and materials for containment and cleaning up	• Contain spillage, soak up with non-combustible absorbent material and transfer to a container for disposal according to local / national regulations

SECTION 7: Handling and storage

Advice on safe handling	• Avoid contact with skin and eyes. • Avoid formation of dust and aerosols.
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Conditions for safe storage	• Provide appropriate exhaust ventilation at places where dust is formed.
	• Keep away from source of ignition
	• Take measures to prevent the buildup of electrostatic charge
	• Store in cool place. Keep container tightly closed in a dry and well-ventilated place
	• Storage class (TRGS 510): 13: Non Combustible Solids

SECTION 8: Exposure controls/personal protection

Control parameters

Exposure controls

Engineering measures	: Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level.
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Individual protection measures, such as personal protective equipment

General information	: An eye wash bottle must be available at the work site. Provide access to washing facilities including soap, skin cleanser and fatty cream
Eye/face protection	: Safety glasses with side-shields conforming to EN166 Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

Skin protection	: Wear suitable gloves. When handling hot material, use heat resistant gloves. The selected protective gloves have to satisfy the specifications of EU Directive 2016/425 and the standard EN 374 derived from it.
Respiratory Protection	: For nuisance exposures use type P95 (US) or type P1 (EU EN 143) particle respirator. For higher level protection use type OV/AG/P99 (US) or type ABEK-P2 (EU EN 143) respirator cartridges. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).
Hygiene measures	: Handle in accordance with good industrial hygiene and safety practice. Do not eat, drink or smoke when using the product. Wash at the end of each work shift and before eating, smoking and using the toilet. Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reuse. Keep away from food, drink and animal feeding stuffs.
Environmental Controls	Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels. Do not contaminate water sources or sewer.

SECTION 9: Physical and chemical properties

Information on basic physical and chemical properties

Physical state : Liquid

Color	: Colorless
Odor	: Mild
Odor Threshold	: No data available.
pH	: No data available
Freezing Point	:
Boiling Point	:
Flash Point	: 110
Density	: 1030.7 kg/m ³ (at 25 °C)
Dynamic Viscosity	: 3.041 mPa-sec (at 25 °C)
Solubility in Water	:
Partition coefficient	:
Autoignition Temperature	:
Decomposition Temperature	:
Explosive properties	: Not classified
Oxidizing properties	: Not classified

SECTION 10: Stability and reactivity

Reactivity	: None reasonably foreseeable
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: None known.
Conditions to avoid	: Heating in air. & Keep away from flames and sparks.
Incompatible materials	: Strong oxidizing agents
Hazardous decomposition products	:

SECTION 11: Toxicological information

Acute oral toxicity	N.A
Acute inhalation toxicity	
Acute dermal toxicity	

Skin corrosion/irritation	
Serious eye damage/eye irritation	
Respiratory or skin sensitization	
Germ cell mutagenicity	
Carcinogenicity	
Reproductive toxicity	
Repeated dose toxicity	
Aspiration toxicity	

SECTION 12: Ecological information

Ecotoxicity	N.A
Toxicity to fish	
Toxicity to daphnia and other aquatic invertebrates	
Toxicity to algae	
Persistence and degradability	
Biodegradability	
Biochemical Oxygen Demand (BOD)	
Chemical Oxygen Demand (COD)	
Bioaccumulative potential	
Bioaccumulation	
Mobility in soil	

SECTION 13: Disposal considerations

Disposal methods	: Recover and reclaim or recycle, if practical. Dispose of this material and its container to hazardous or special waste collection point. Dispose of waste at an appropriate treatment
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	and disposal facility in accordance with applicable laws and regulations, and product characteristics at time of disposal. Do not discharge into drains, water courses or onto the ground
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